

14.5.15 (E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam

April/May 2015

Max. Marks: 100

Class: SE

Name of the Course: Object Oriented Programming Methodology

Course Code: UCEC302

Duration: 3hrs

Semester: III

Branch: COMP

Instructions:

- (1) All Questions are Compulsory.
- (2) Assume suitable data if necessary.

Question No.		Max. Marks
Q1(a)	Elaborate the use of System.arraycopy() with an example.	[05]
	OR	
	What is an array, explain with an example different ways to declare and initialize an array.	
Q1(b)	Write short notes on wrapper classes.	[05]
Q1(c)	Explain different types of inheritance with example.	[05]
Q1(d)	Elaborate the use of toString() method with example.	[05]
Q2(a)	Write a program to display area of square and rectangle. Make use of interface to define templates of methods to be there in the derive classes.	[10]
	OR	
	What is an abstract class? How it is different from interface explain with an example.	
Q2(b)	What are the benefits of packages? What are the Java API packages? Write a program to demonstrate the creation and use of a package.	[10]
Q3(a)	What is an exception? How exceptions are handled in JAVA, explain by giving examples.	[10]
Q3(b)	Explain method overloading and method overriding with suitable example.	[10]

Q4 (a)	With the help of suitable example explain multithreading in terms of following:- (i) Creating threads, extending the thread class. (ii) Life-cycle of Thread. OR With the help of suitable example explain synchronization of threads.	[10]
Q4 (b)	What is an applet? Explain life cycle of an applet and write an applet program to draw circle, rectangle and line.	[10]
Q5 (a)	Write a program in JAVA that computes the sum of a list of integers that is supplied by a user. The end of data signaled by the value -999. This value is used only as a flag and not used in sum. OR Write a program to accept a number from command line and to check whether it is Armstrong Number or Not.	[10]
Q5 (b)	Assume that a bank maintains two kinds of accounts for customers, one called as savings account and the other as current account. The savings account provides compound interest and withdrawal facilities but no cheque book facility. The current account provides cheque book facility but no interest. Current account holders should also maintain a minimum balance and if the balance falls below this level, a service charge is imposed. Class account stores customer name, account number, and the type of account. Include member functions :- • Accept deposit from a customer and update the balance. • Display the balance. • Compute and deposit interest. • Permit withdrawal and update balance. • Check for minimum balance, impose penalty and update the balance. Draw class diagram for the above scenario. OR Implement class account, current account and savings account for the above mentioned scenario.	[10]